Set 11 at Daan — the front window where the cab would be.

← Back to Rolling Stock

VAL256

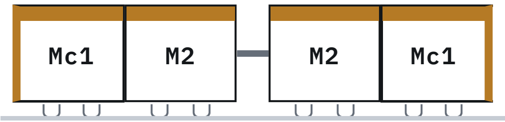
The French fleet that opened the network in 1996 — and spent eighteen months in a depot having its brain replaced while the line it built ran without it.

BR WENHU LINE [/RAIL/LINES/WENHU-LINE/] 文湖線

BUILDER	FAMILY	OPERATOR	LINE
Matra Transport, with GEC Alsthom [1]	VAL	Taipei Rapid Transit Corporation [/rail/operators/trtc/]	Wenhu [/rail/lines/wenhu-line/]
BUILT	**IN SERVICE**	**DEPOT**	
1990—1993 [1]	28 March 1996 — [6]	Muzha [/rail/depots/muzha-depot/] and Neihu [/rail/depots/neihu-depot/] [7]	

Stub page. This entry is an outline. Figures marked *TBC* still need to be checked against a primary source before publication.

Formation



4 CARS · 2 UNITS OF 2 · ONE UNIT COUPLING CAB

A train delivered in the early 1990s is still running the line, and almost nothing tells it what to do any more.

The VAL256 sets were built by Matra with GEC Alsthom between 1990 and 1993[1] and entered service with the Muzha Line on 28 March 1996[6]. The `256` is the car width in centimetres — 2,560 mm — and it is the wider of the two standard VAL profiles, the other being the 2,060 mm VAL206 used at Lille[4].

The re-signalling

The conversion did not precede the Neihu [/rail/stations/br19/] extension — it was triggered by it, and the sequence is the story:

DATE	EVENT
4 July 2009	The Neihu extension opens. The line switches to CITYFLO 650, the Matra system is cut off after thirteen years, and the VAL256 fleet **comes out of service that day** to begin conversion[1] [5]
26 December 2010	Converted VAL256 units return to passenger service, running alongside the Innovia fleet[1]

So for nearly eighteen months the whole 24-station line ran on the Bombardier fleet alone, while the trains that had opened the railway sat in a depot having their control system replaced. Why the line was re-signalled at all — and why by a competitor of the system's owner — is told on The Matra dispute [/rail/history/matra-dispute/].

The original equipment was a fixed-block ATC/ATO system derived from the MAGGALY technology on Lyon Metro line D[2]; what replaced it is a moving-block communications-based system.

Design

VAL vehicles run on rubber tyres over concrete running surfaces, and are guided from the **sides**, not the centre: pairs of horizontal wheels bear outward against lateral guide bars — H-section steel, set roughly 200 mm above the running surface on either side of the vehicle — and those same bars carry the 750 V supply, which is why a VAL guideway needs no separate conductor rail [3] [4].

VAL's own successor went the other way. The NeoVal generation Siemens announced in 2006 *is* centrally guided, on a single rail similar to Translohr's [4]. Central guidance is VAL's future, not its past, and Taipei has the past.

The formation is not fixed

Cars are delivered and counted in pairs — DORTS bought 51 of them under contract CC350 [8] — and two pairs make a four-car train. But **the pairs are not permanently married into sets**: zh.wikipedia describes this fleet as 「非固定編組」, a non-fixed formation, with a train made up of four consecutively-numbered cars rather than of one unchanging four-car unit [9].

That detail does more work than it looks. TRTC `[/rail/operators/trtc/]`'s own announcement of the seat-removal programme counts **25 four-car trains** [10], and 25 trains is 50 pairs against a fleet of 51. If the sets were fixed, that odd pair would be a permanent orphan needing an explanation. If the formation is not fixed, there is nothing to explain — the cars run as 25 trains [10] and which cars those are is a maintenance question rather than a fact about the fleet [9].

Confidence is Medium and the page does not overstate it. The pair counts and the train count are both primary; 非固定編組 is one encyclopedia, and no manufacturer or operator document found says it. It is also the reading that resolves the line-level arithmetic problem `[/rail/lines/wenhu-line/]` without requiring a train formed from both fleets, which is a reason to want it to be true and therefore a reason to mark it carefully.

Fleet history

Ordered for the Muzha Line and in service from its opening on 28 March 1996, originally working **BR01** through **BR12** only. Since December 2010 the fleet has run the full route alongside the Innovia APM 256 (C370) `[/rail/rolling-stock/innovia-apm-256-c370/]`.

Delivery dates by unit, and the scope of the re-signalling work on each set, are still to be confirmed.

Depot allocation

Maintained at Muzha Depot `[/rail/depots/muzha-depot/]`, beside **BR01**, which also stables Innovia sets [7].

Corrections

- **This page previously said the fleet was re-signalled ahead of the 2009 extension.** That inverts the sequence: the conversion began the day the extension opened and the fleet was out of service until December 2010[1].
- **This page previously said VAL vehicles are guided by horizontal wheels bearing on a central rail.** They are side-guided[3][4]. The likely origin of the error is the NeoVal succession — the *later* generation is centrally guided, and the two were run together here.

Specifications

Car width	2.56 m[1]
Car length	13.78 m[1]
Car height	3.53 m[1]
Cars per train	4 [1]
Empty weight	TBC t
Design maximum speed	80 km/h[1]
Operating speed	70 km/h[1]
Seated capacity	20 per car (AW2) [1]
Standing capacity	94 per car (AW2) [1]
Train capacity	440 [11]
Doors per car	4 2 per side[1]
Power supply	750 V DC[1]
Traction	GEC Alsthom chopper DC[1]
Motors per car	2 [1]
Braking	Regenerative and disc [1]
Guidance	Side guide bars [3]
Signalling	CITYFLO 650 since 2010[1]
Automation	GoA4 (UTO) [5]
Fleet size	102 cars[1]
Pairs	51 [8]
Tyre replacement interval	TBC

References

3 of 11 sources on this page are primary — published by the operator, the builder or government.

1 Taipei Metro VAL256 electric multiple unit [https://zh.wikipedia.org/zh-tw/台北捷運VAL256型電聯車]

台北捷運VAL256型電聯車
SECONDARY 維基百科（Chinese Wikipedia） accessed 2026-08-06
archived [https://web.archive.org/web/20221006051618/https://zh.wikipedia.org/zh-tw/台北捷運VAL256型電聯車]
The fullest specification found for this fleet. The article carries no reference section, so every figure taken from it is one source deep — corroborated below by the English article, which is independently worded.

2 Taipei Metro VAL256 [https://en.wikipedia.org/wiki/Taipei_Metro_VAL256]
SECONDARY English Wikipedia accessed 2026-08-06
archived [https://web.archive.org/web/20260511224742/https://en.wikipedia.org/wiki/Taipei_Metro_VAL256]
Agrees with the Chinese article on every dimension, capacity, speed and date. Adds that the original signalling derived from the MAGGALY system on Lyon Metro line D.

3 Lille VAL — project profile [https://www.railway-technology.com/projects/lille_val/]
SECONDARY Railway Technology accessed 2026-08-06
archived [https://web.archive.org/web/20260807044500/https://www.railway-technology.com/projects/lille_val/]
Describes the guideway directly — running surfaces, lateral H-section guide bars about 200 mm above the running surface, and those bars doubling as the 750 V supply. Trade press, not a manufacturer drawing; a Siemens or Matra engineering document would settle it properly.

4 Véhicule Automatique Léger [https://en.wikipedia.org/wiki/V%C3%A9hicule_Automatique_L%C3%A9ger]
SECONDARY English Wikipedia accessed 2026-08-06
archived [https://web.archive.org/web/20260807045415/https://en.wikipedia.org/wiki/V%C3%A9hicule_Automatique_L%C3%A9ger]
Independent corroboration of side guidance — "pairs of horizontal tires to provide lateral guidance", with power "collected by shoes from the guidebars". Also records that the later NeoVal moved to a single central rail, which is the most likely origin of the central-rail error this page used to carry.

5 Wenhu line [https://zh.wikipedia.org/zh-tw/文湖線]
文湖線
SECONDARY 維基百科（Chinese Wikipedia） accessed 2026-08-06
archived [https://web.archive.org/web/20250714045049/https://zh.wikipedia.org/zh-tw/文湖線]

6 Wenshan—Neihu Line [https://www.dorts.gov.taipei/cp.aspx?n=DBAC040496EFAB94]
文山內湖線
PRIMARY Taipei City Government, Department of Rapid Transit Systems (DORTS) accessed 2026-08-06
archived [https://web.archive.org/web/20260209210832/https://www.dorts.gov.taipei/cp.aspx?n=DBAC040496EFAB94]

7 Muzha Depot [https://zh.wikipedia.org/zh-tw/木柵機廠]
木柵機廠
SECONDARY 維基百科（Chinese Wikipedia） accessed 2026-08-06
archived [https://web.archive.org/web/20220520160336/https://zh.wikipedia.org/zh-tw/木柵機廠]

8 How many trains were procured for each line of the Taipei metro network? [https://www.dorts.gov.taipei/News_Content.aspx?n=2A66A485FACB0D5B&s=C8602F8588914E91]
台北都會區大眾捷運系統路網中，各路線所採購之列車數為何？
PRIMARY Taipei City Government, Department of Rapid Transit Systems (DORTS) accessed 2026-08-07
archived [https://web.archive.org/web/20260807044541/https://www.dorts.gov.taipei/News_Content.aspx?n=2A66A485FACB0D5B&s=C8602F8588914E91]
The builder's own procurement table: 木柵線 CC350 51對電聯車（102輛）. The contract number and the pair count.

9 Taipei Metro rolling stock [https://zh.wikipedia.org/zh-hant/臺北捷運列車]
臺北捷運列車
SECONDARY 維基百科（Chinese Wikipedia） accessed 2026-08-07
archived [https://web.archive.org/web/20260807045301/https://zh.wikipedia.org/zh-hant/%E8%87%BA%E5%8C%97%E6%8D%B7%E9%81%8B%E5%88%97%E8%BB%8A]
The only source found for this fleet running 「非固定編組」 — a non-fixed formation, four consecutively-numbered cars making a train rather than a permanent four-car unit. It is the claim the site's formation notation used to contradict, and no manufacturer or operator document has been found either way.

10 Taipei Metro turns space magician: after the Bombardier fleet, the Wenhu line's Matra trains get an interior optimisation [https://www.metro.taipei/News_Content.aspx?n=30CCEFD2A45592BF&sms=72544237BBE4C5F6&s=86F98235BC105C44]
臺北捷運化身空間魔術師！繼龐巴迪列車改善後 文湖線馬特拉列車空間優化
PRIMARY Taipei Rapid Transit Corporation (news release) accessed 2026-08-06
archived [https://web.archive.org/web/20250723182554/https://www.metro.taipei/News_Content.aspx?n=30CCEFD2A45592BF&sms=72544237BBE4C5F6&s=86F98235BC105C44]
The operator's own announcement of the seat-removal programme, and the source for this fleet forming 25 four-car trains.

11 The whole Wenhu line to be renewed — new trains gangwayed across four cars, capacity up 30% to 574 [https://www.ettoday.net/news/20260507/3162180.htm]
文湖線將全線更新　新列車採「四節貫穿」、載客量增30%至574人
SECONDARY ETtoday 新聞雲 , 7 May 2026 accessed 2026-08-06
archived [https://web.archive.org/web/20260807045443/https://www.ettoday.net/news/20260507/3162180.htm]
Gives 440 as the current Matra train capacity, as the comparison for the replacement programme. Treat as provisional — it is a figure quoted in a council exchange reported in news, not a published specification, and it does not reconcile exactly with 114 per car.

Last updated: 2026-08-06

Station names, codes and sequence from TDX Transport Data eXchange, Ministry of Transportation and Communications, operators TRTC, NTMC, and TYMC. Government open data. Retrieved 2026-08-10; source last updated 2020-01-31. The 12 Sanying station records absent from TDX are sourced directly to New Taipei Metro and government publications on their pages. Provenance.